

Universal Bound on Energy Cost of Bit Reset in Finite Time

Yi-Zheng Zhen^{1,2}, Dario Egloff^{3,4}, Kavan Modi^{5,1} and Oscar Dahlsten^{1,*}

¹*Shenzhen Institute for Quantum Science and Engineering and Department of Physics, Southern University of Science and Technology, Shenzhen 518055, China*

²*Hefei National Laboratory for Physical Sciences at Microscale and Department of Modern Physics, University of Science and Technology of China, Hefei 230026, China*

³*Institute of Theoretical Physics, Technische Universität Dresden, D-01062 Dresden, Germany*

⁴*Max Planck Institute for the Physics of Complex Systems, Nöthnitzer Strasse 38, 01187 Dresden, Germany*

⁵*School of Physics and Astronomy, Monash University, Clayton, Victoria 3800, Australia*



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We consider how the energy cost of bit reset scales with the time duration of the protocol. Bit reset necessarily takes place in finite time, where there is an extra penalty on top of the quasistatic work cost derived by Landauer. This extra energy is dissipated as heat in the computer, inducing a fundamental limit on the speed of irreversible computers. We formulate a hardware-independent expression for this limit in the framework of stochastic processes. We derive a closed-form lower bound on the work penalty as a function of the time taken for the protocol and bit reset error. It holds for discrete as well as continuous systems, assuming only that the master equation respects detailed balance.

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Bit reset, the setting of an unknown bit to a fixed value, is an elementary operation in irreversible computation. Landauer's principle states that bit reset must cost at least $k_B T \ln 2$ amount of energy, where k_B is Boltzmann's constant and T the temperature of the ambient environment, i.e., the computer [1–3]. The energy cost of binary transistor switching is extrapolated to reach this Landauer limit around 2035 [4]. This cost is a crucial concern: in the next ten years, digital information processors are expected to consume 1/5 of the world's electricity [5]. Moreover, the energy expended is dissipated as heat into the computer and computers are already speed constrained by power dissipation tolerance limits, posing a key problem in continuing Moore's law indefinitely [6]. Apart from this technological importance, Landauer's principle is also a focal point concerning the role of information in thermodynamics [7–11] and has also been extended into the quantum regime, see, e.g., [12–18].

To achieve the $k_B T \ln 2$ limit, quasistatic protocols are required and often assumed possible [7,8,10]. However, real-life bit reset [19–25] takes place in finite time. This necessitates a possibly dramatically higher work cost than $k_B T \ln 2$: there is a finite-time work penalty to bit reset. Understanding this penalty is crucial to the energy efficiency of computers in approaching the Landauer limit of irreversible computing. Generally, the scaling of power dissipation within a time duration τ has been argued to be inversely linear in the long-time limit for any stochastic process [26] and, in the slow-driving case, for classical [27] or quantum systems [15,16]. For bit reset, this scaling also matches theoretical [28–32] and experimental results

[19,21] on a mesoscopic system immersed in a double-well potential. Particularly, the optimal reset protocol for the double-well case shows that the additional work (beyond the Landauer limit) scales as $1/\tau$ for cases such as vanishing reset error [31] or local equilibrium [32]. Meanwhile, a theoretical model for the reset of a qubit indicates a more benign scaling [33]. The tension between these results, as well as the multitude of possible hardware implementations of bits, including colloidal systems and fluctuating RC circuits [29,33–38], motivates a search for a universal Landauer's principle that holds in finite time regardless of the physical implementation. This will help us understand how far one can minimize the power dissipation, in principle, by altering the hardware, whether with complementary metal-oxide–semiconductor or alternative platforms [39].

In this Letter, we analytically derive a universal lower bound for the finite-time work penalty in bit reset. The bound applies to all physical models as long as detailed balance is respected and, thus, encompasses previous results focused on specific models such as those depicted in Figs. 1(a) and 1(b). The bound also holds for both the total time region and the finite bit reset error and, thus, can be viewed as a renewed Landauer's principle for finite as well as infinite time. The proof of the bound implies that the work penalty is optimized if one adopts a truly two-level system, such as a spin. We further apply the bound to induce an upper limit on the information throughput of irreversible computers. Our results will aid nanoelectronics research on alternative hardware and in extending information thermodynamics to finite-time nonequilibrium regimes.

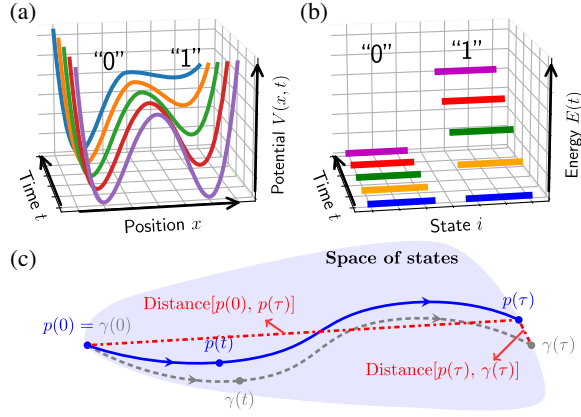


FIG. 1. Our analysis applies to a wide range of bit-reset scenarios, including: (a) a double-well potential $V(x, t)$ which is gradually turned into a single well, (b) a system with two energy levels where one level is lifted gradually. (c) During the finite-time bit reset, the system state $p(t)$ lags behind the thermal state $\gamma(t)$, and we will show that two distances are closely related to the minimal work cost.

Stochastic thermodynamics.—We briefly introduce common definitions for stochastic thermodynamics using discrete states, though the results straightforwardly apply in the continuous case (see [40,41] and the Supplemental Material [42]). Denote the state of a physical system by $i \in \{1, 2, \dots, N\}$ with an associated energy E_i . We use $p(t) = [p_1(t), \dots, p_N(t)]$ as the distribution of system states at time t . A key assumption is that when the system is in contact with a heat bath at temperature T , this distribution undergoes a stochastic evolution toward the thermal state γ , wherein $\gamma_i = e^{-\beta E_i}/Z$ with $\beta = 1/k_B T$ and the partition function $Z = \sum_i e^{-\beta E_i}$. The evolution is modeled by a Markovian master equation

$$\frac{d}{dt} p_i(t) = \sum_j \Gamma_{ij}(t) p_j(t). \quad (1)$$

Here, $\Gamma_{ij}(t)$ is the transition rate from state j to state i which satisfies $\Gamma_{ij}(t) \geq 0$ for all $i \neq j$ and $\sum_i \Gamma_{ij}(t) = 0$ for any j due to conservation of probability. We also assume detailed balance $\Gamma_{ij}(t) \gamma_j(t) = \Gamma_{ji}(t) \gamma_i(t)$ for all $i \neq j$, which is a common condition satisfied by a wide class of nonequilibrium processes [43].

The system is allowed to interact with a work reservoir, which “drives” the system [44]. Then, any change in internal energy $U = \sum_i p_i E_i$ is associated with work (due to the work reservoir) and heat (due to the heat bath), i.e., $dU = \sum_i E_i dp_i + \sum_i p_i dE_i = dQ + dW$. For a process taking time τ , the heat exchanged and work input are then $Q(\tau) = \int_0^\tau \dot{Q} dt$ and $W(\tau) = \int_0^\tau \dot{W} dt$, respectively. When the driving is very slow, the system remains in the instantaneous thermal state at all times. Such a process is called quasistatic with associated work $W_{qs}(\tau)$ [45].

The W_{qs} can be determined or approximated experimentally either by slowing down the protocol until the work cost is almost constant under further slowing, or in principle, from fast experiments by means of the Jarzynski equality (which follows from detailed balance) [46]. The difference between the actual work cost and W_{qs} can be termed the work penalty

$$W_{pn}(\tau) = W(\tau) - W_{qs}(\tau). \quad (2)$$

The quasistatic entropy change $\Delta S(\tau) = S(\tau) - S(0)$, where $S = -k_B \sum_i p_i \ln p_i$, is $Q(\tau)/T$, and any additional entropy change is termed the entropy production [40,41]: $\Sigma(\tau) = \Delta S(\tau) - Q(\tau)/T \geq 0$. It is often convenient to use the (non-negative) entropy production rate

$$\frac{d}{dt} \Sigma = \frac{k_B}{2} \sum_{i \neq j} (\Gamma_{ij} p_j - \Gamma_{ji} p_i) \ln \frac{\Gamma_{ij} p_j}{\Gamma_{ji} p_i}, \quad (3)$$

for the case of detailed balance.

Generic bit reset.—Now, we model the bit reset process with stochastic thermodynamics. The bit is associated with two logical states “0” and “1.” The bit is reset (or “erased”) if it is set to 0 regardless of the initial state. It is normally assumed that either logical state is initially equally likely. As can be seen from Fig. 1(a), logical states may be associated with sets of underlying microstates, Ω_0 and Ω_1 such that

$$P_a^{\text{bit}}(t) = \sum_{i \in \Omega_a} p_i(t), \quad \forall a \in \{0, 1\}. \quad (4)$$

Equation (4) defines the coarse-grained [47] bit distribution $P^{\text{bit}}(t)$. The coarse-grained thermal state can be similarly defined as γ^{bit} with $\gamma_a^{\text{bit}} = \sum_{i \in \Omega_a} \gamma_i$ for $a \in \{0, 1\}$.

By employing interactions with a heat bath and with a work reservoir [48], a reset protocol monotonically increases the energy differences between $\{E_i | i \in \Omega_0\}$ and $\{E_j | j \in \Omega_1\}$, while the system undergoes thermalization. As depicted in Fig. 1(a) and 1(b), P_1^{bit} decreases leading to a final distribution $P^{\text{bit}}(\tau) = [1 - \epsilon, \epsilon]$, where $\epsilon = P_1^{\text{bit}}(\tau)$ is the reset error.

Landauer’s principle states that the minimal work cost for such a protocol (with $\epsilon \rightarrow 0$) is $W_{qs} = k_B T \ln 2$. This can be shown to follow from the non-negativity of entropy production [cf. Eq. (6) below]. Now, we proceed to bound the work penalty for finite time and nonvanishing reset error.

Main result: Bound on work penalty.—For any bit reset protocols within the framework described above, if initially the system is fully thermalized and $P^{\text{bit}}(t=0) = [1/2, 1/2]$, the work penalty satisfies

$$\beta W_{\text{pn}}(\tau) \geq D_\epsilon(\tau) + \frac{(1 - 2\epsilon)^2}{\langle \mu \rangle_\tau \tau}, \quad (5)$$

where $D_\epsilon(\tau) = D[P^{\text{bit}}(\tau) \parallel \gamma^{\text{bit}}(\tau)]$ is the relative entropy between the coarse-grained bit state with coarse-grained thermal state at the final time and $P_1^{\text{bit}}(\tau) = \epsilon$. The reset error ϵ can be demanded as a boundary condition, or it can be treated as a parameter that depends on $\langle \mu \rangle_\tau$ for allowed maximal energies. $\langle \mu \rangle_\tau$ is a measure of how strong the thermalization is, as discussed later around Eq. (12). The proof has three steps (see Sec. I of the Supplemental Material [42]).

Step I: Work penalty from relative entropy: For the (fine-grained) physical systems, the work penalty of any finite-time protocol can be expressed as

$$W_{\text{pn}}(\tau) = k_B T \Delta D[p \parallel \gamma] + T \Sigma(\tau), \quad (6)$$

where $\Delta D[p \parallel \gamma] = D[p(\tau) \parallel \gamma(\tau)] - D[p(0) \parallel \gamma(0)]$ is the change of relative entropy $D[p \parallel \gamma] = \sum_i p_i \ln(p_i/\gamma_i)$ (see, e.g., Ref. [49]). This can be understood from the perspective of information geometry [50,51]. As $p(t)$ always chases after $\gamma(t)$ due to thermalization [see Fig. 1(c)], we examine the rate of change in “distance” between them

$$\dot{D}[p(t) \parallel \gamma(t)] = \dot{p} \partial_p D[p \parallel \gamma] + \dot{\gamma} \partial_\gamma D[p \parallel \gamma], \quad (7)$$

where $\dot{p} \partial_p D[p \parallel \gamma] = \sum_i \dot{p}_i \partial_{p_i} D[p \parallel \gamma]$ and $\dot{\gamma} \partial_\gamma D[p \parallel \gamma] = \sum_i \dot{\gamma}_i \partial_{\gamma_i} D[p \parallel \gamma]$. As proved in Sec. I(A) of the Supplemental Material [42], the first term is identical to entropy production rate $\dot{\Sigma}$ up to a factor $-k_B$, while the second term is exactly the time derivative of βW_{pn} . By integrating the equation $\dot{D}[p \parallel \gamma] = -\dot{\Sigma}/k_B + \beta \dot{W}_{\text{pn}}$, Eq. (6) is obtained.

Step 2: Coarse-graining relative entropy, entropy production, and dynamics: The underlying state and dynamics induce corresponding coarse-grained states and dynamics, as described above. For general bit reset protocols, the system is initially in a thermal state such that $\Delta D[p \parallel \gamma] = D[p(\tau) \parallel \gamma(\tau)]$ in Eq. (6). First, we show that $D[p(\tau) \parallel \gamma(\tau)]$ can be lower bounded by its coarse-grained counterpart (see Sec. I(B) of the Supplemental Material [42])

$$D[p(\tau) \parallel \gamma(\tau)] \geq D[P^{\text{bit}}(\tau) \parallel \gamma^{\text{bit}}(\tau)] \equiv D_\epsilon(\tau). \quad (8)$$

To lower bound entropy production, we require the coarse-grained dynamics. As a hidden-Markov model, the coarse-grained dynamics can be described by the master equation $\dot{P}_a^{\text{bit}}(t) = \sum_a \Gamma_{aa}^{\text{bit}}(t) P_a^{\text{bit}}(t)$ for $a \in \{0, 1\}$, with a transition rate $\Gamma_{aa}^{\text{bit}}(t)$ that depends on the underlying microstates [47], which, in turn, are uniquely determined by the initial microstate for the given dynamics. We define a coarse-grained entropy production rate [cf. Eq. (3)]

$$\frac{d}{dt} \Sigma^{\text{bit}} = \frac{k_B}{2} \sum_{a \in \{0,1\}} (\Gamma_{aa}^{\text{bit}} P_a - \Gamma_{aa}^{\text{bit}} P_a) \ln \frac{\Gamma_{aa}^{\text{bit}} P_a}{\Gamma_{aa}^{\text{bit}} P_a}. \quad (9)$$

Note that $\dot{\Sigma}^{\text{bit}}$ is experimentally accessible via measuring Γ_{aa} following the coarse-grained trajectories, as in, e.g., [19,22,52,53]. We prove in Sec. I(B) of the Supplemental Material [42] that this $\Sigma^{\text{bit}}(\tau)$ is non-negative and is a lower bound to the exact $\Sigma(\tau)$ of the system, i.e.,

$$\Sigma(\tau) \geq \Sigma^{\text{bit}}(\tau). \quad (10)$$

Then, combining Eqs. (6), (8), and (10), the work penalty of bit reset is always lower bounded by

$$W_{\text{pn}}(\tau) \geq k_B T D[P^{\text{bit}}(\tau) \parallel \gamma^{\text{bit}}(\tau)] + T \Sigma^{\text{bit}}(\tau). \quad (11)$$

The bound in Eq. (11) is tight in the case of local equilibrium [32,47], wherein the probabilities of microstates conditional on the bit value are proportional to those of a thermal state (see Sec. I(C) of the Supplemental Material [42]). This also strongly suggests that two-level systems outperform mesoscopic systems in terms of work penalty in general: for any coarse-grained master equation, there exists a qubit evolution such that the work penalty lower bound is saturated. We have not allowed for the qubit to have quantum coherence here; there are reasons to think it does not help in bit reset protocols [33].

Step 3: Entropy production time scaling via speed limit: First, we turn our attention to the τ dependence of the second term of Eq. (11) by extending the recently discovered classical speed limit [54] approach to the coarse-grained case. Speed limits were first proposed to operationalize the time-energy uncertainty relation [55] and, since then, have found utility in many facets of quantum and classical physics [56–68]. In essence, the changes to the state of a system must occur in finite time, which depends on the distance between the initial and the final states and the speed, i.e., the system’s energetics [69].

We show that the coarse-grained master equation can always be mapped to a partial swap model $\dot{P}^{\text{bit}}(t) = \mu(t)[P^{\text{bit}}(t) - P^{\text{st}}(t)]$, with a parameter $\mu(t) = \sum_a \Gamma_{aa}^{\text{bit}}(t)$ describing the swap rate during relaxation and $P^{\text{st}}(t)$ the coarse-grained stationary state [70] (see Sec. I(D) of the Supplemental Material [42]). Using this model, we prove the speed limit for the coarse-grained system,

$$\tau \geq k_B \frac{L[P^{\text{bit}}(\tau), P^{\text{bit}}(0)]^2}{\langle \mu \rangle_\tau \Sigma^{\text{bit}}(\tau)}, \quad (12)$$

where $\langle \mu \rangle_\tau = \tau^{-1} \int_0^\tau dt \mu(t)$ is the time averaged swap rate, and $L[P^{\text{bit}}(\tau), P^{\text{bit}}(0)] = \sum_a |P_a^{\text{bit}}(\tau) - P_a^{\text{bit}}(0)|$ is the 1-norm distance (see proof in Sec. I(E) of the Supplemental

Material [42]). The “speed” here is, thus, $\langle \mu \rangle_\tau \Sigma^{\text{bit}}(\tau)$. As was the case for $\dot{\Sigma}^{\text{bit}}$, $\langle \mu \rangle_\tau$ can also be obtained by following the coarse-grained trajectories in experiments via $\mu(t) = \Gamma_{01}^{\text{bit}}(t) + \Gamma_{10}^{\text{bit}}(t)$. Moreover, in Sec. I(F) of the Supplemental Material [42] we describe how $\langle \mu \rangle_\tau$ can be derived exactly or upper bounded given a master equation. We remark that the bound is tight in the quasistatic limit, as one can see, by inspection, that both sides of Eq. (12) go to 0 in that limit. It also performs well in finite times in the example investigated in Fig. 2 below.

Finally, using Eqs. (11) and (12) while substituting the initial condition $P^{\text{bit}}(0) = [1/2, 1/2]$ and final condition $P^{\text{bit}}(\tau) = [1 - \epsilon, \epsilon]$ for a generic bit reset protocol, the penalty bound Eq. (5) is deduced.

Relative entropy time scaling.—There are reasons to think that the relative entropy term will, in general, decay exponentially in τ . For example, we derive a general lower bound (in Sec. II of the Supplemental Material [42]) which has exponential decay for the case of local equilibrium as well as more generally for short times. For the latter case, it takes the form $D_\epsilon(\tau) \geq e^{-\langle \mu \rangle_\tau \tau} (2e^{-\langle \mu \rangle_\tau \tau} - 1) D[\gamma^{\text{bit}}(0) \parallel \gamma^{\text{bit}}(\tau)]$. This is in contrast to the speed limit lower bound $\Sigma^{\text{bit}}(\tau)$, which has inverse linear scaling in τ . To investigate this in more depth, now, we specialize to a typical reset protocol in a two-level system.

Discrete level-shifting protocol in a two-level system.—As we have shown in Eq. (11), a two-level system may optimize the work penalty. For such a system, e.g., Fig. 1(b), coarse-grained and fine-grained states coincide, such that the evolution of the system is described by the partial swap model $\dot{P}(t) = \mu[\gamma(t) - P(t)]$, where we have assumed $\mu(t) = \mu$ for simplicity. This model is a simplified collisional model for thermalization [33,71], which has been used to describe relaxation of nuclear magnetic resonance qubits.

A typical reset protocol is to shift one energy level at a constant rate, as discussed in [33]. The two energy levels are both zeros initially, and then, the second energy level is shifted to a maximal energy E_{max} in N steps: for each step, E_1 is suddenly lifted with an increment E_{max}/N , and then, the system thermalizes for a time τ/N . Physically, this assumes the experimentalist can change the control fields significantly faster than the thermalization rate. We prove that $D_\epsilon(\tau)$ in Eq. (5) must have an exponential scaling (see Sec. III(B) of the Supplemental Material [42] for the proof),

$$D_\epsilon(\tau) \leq e^{-\mu\tau/N} D[\gamma(0) \parallel \gamma(\tau)]. \quad (13)$$

Figure 2 gives numerical simulations showing the validity of Eq. (5) and its tightness. Two qualitatively different cases are examined: that of fixed maximal energy level E_{max} , and that of fixed reset error ϵ . For both cases, the bound in Eq. (5) restricts the work penalty closely and the bound becomes tighter as the finite time gets longer.

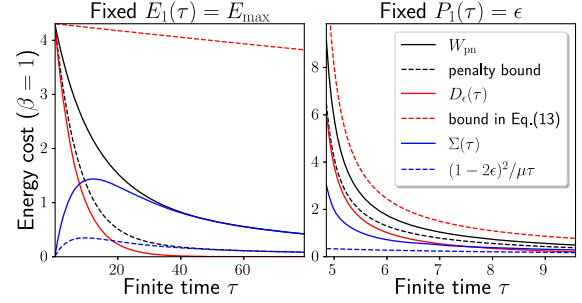


FIG. 2. Time scaling of work penalty, relative entropy, and entropy production for the discrete level-shifting protocol. In the simulation, $N = 100$, $\mu = 0.1$, $\beta = 1$, $E_{\text{max}} = 10$ (left), and $\epsilon = 0.25$ (right).

The relative entropy and entropy production are upper and lower bounded in Eqs. (13) and (12), respectively. We can observe two time regimes: for short times $D_\epsilon(\tau)$ dominates, and $W_{\text{pn}}(\tau)$ decays exponentially, and for long times, $\Sigma(\tau)$ dominates such that $W_{\text{pn}}(\tau)$ cannot escape an inverse linear scaling. As we prove in Sec. III(D) of the Supplemental Material [42], the work penalty itself can be bounded by an exponentially decaying function which converges to a positive value. Therefore, an exponential decay of the work penalty (in τ) can, in certain regimes, dominate the known inverse linear scaling. To understand the boundaries between the regions where $D_\epsilon(\tau)$ or $\Sigma(\tau)$ dominate the work penalty, in Fig. 3, we show the performance of the protocol in the entire region of E_{max} and ϵ . Except for the region where the protocol fails (region III), i.e., where $\epsilon < \gamma_1(E_{\text{max}})$, the work penalty (finite time) increases (decreases) as both E_{max} and ϵ grow. Roughly speaking, when ϵ is large, one can always set E_{max} such that W_{pn} is in an exponential scaling (Region I). When ϵ is small, W_{pn} is mainly in the inverse linear scaling (Region II).

Encompassing previous results.—Our results encompass the overdamped Langevin dynamics treatment of a mesoscopic system in a double-well potential, which is a typical model for bit reset [Fig. 1(a)]. It describes a large

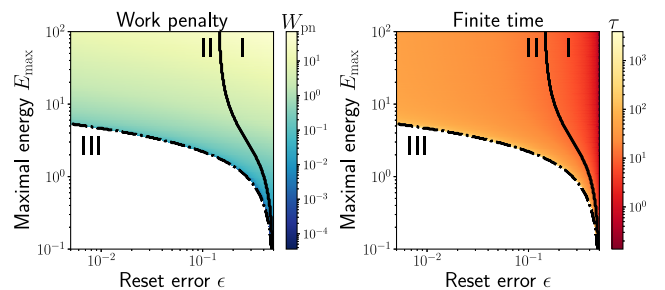


FIG. 3. Domination of $D_\epsilon(\tau)$ or $\Sigma(\tau)$ for the work penalty. Region I: $D_\epsilon(\tau) > \Sigma(\tau)$. Region II: $D_\epsilon(\tau) < \Sigma(\tau)$. Region III: the protocol fails to reset the bit given the error and energy. Solid line: $D_\epsilon(\tau) = \Sigma(\tau)$. Dash-dotted line: $\epsilon = \gamma_1(E_{\text{max}})$. In the simulation, $N = 100$, $\mu = 0.1$, and $\beta = 1$.

class of mesoscopic systems such as a driven particle that also receives random forces from the environment [2,3,28,29,31,32]. Since the underlying Langevin process is Markovian and respects detailed balance, our bound also applies.

In particular, our results can recover the time scaling demonstrated in important works [19,21,31,32]. For experimental works, our results explain the initially observed exponential and, finally, inversely linear work decay [19], and the entire inversely linear scaling for the perfect reset [21]. For theoretical works, our penalty bound is consistent with recent results based on optimal protocol. If the reset error $\epsilon \rightarrow 0$, $D_\epsilon(\tau)$ vanishes, the penalty bound is reduced to $W_{\text{pn}}(\tau) \geq k_B T / \langle \mu \rangle_\tau \tau$, showing a $1/\tau$ scaling [32]. If local equilibrium is assumed, the speed limit bound from Eq. (12) also shows a $1/\tau$ scaling [31]. For generic cases where previous methods fail to give a closed form of work cost, e.g., with a definite reset error, with a specific landscape of potential, or going beyond the Langevin dynamics, Eq. (5) not only applies to give a refined bound with a $1/\tau$ scaling, but also simplifies the derivation of the bound to the question of finding $\langle \mu \rangle_\tau$, which can either be experimentally measured via testing trajectories, or analytically calculated as shown in Sec. IV of the Supplemental Material [42].

Implied bound on information throughput.—As hardware can handle only limited temperatures and the work applied for each reset is dissipated as heat [4], the power dissipation places a limit on how many resets can be done per unit of time and space. Using the notation of [72], for a device switching time τ_{SW} , integration density (number of binary switches per cm^2) n , the information throughput $B = n/\tau_{\text{SW}}$ is restricted by the ratio of binary transition energy E_{bit} and power dissipation growth $P = E_{\text{bit}}B$. As our results concern the minimal work cost of finite-time bit reset in generic hardwares, it leads to a refined restriction on the power dissipation growth, i.e., $P \geq k_B T n [\ln 2 - H_b(\epsilon) + \beta E_{\text{max}} \epsilon + (1 - 2\epsilon)^2 / (\mu \tau_{\text{SW}})] / \tau_{\text{SW}}$, where $H_b(\epsilon) = -\epsilon \ln \epsilon - (1 - \epsilon) \ln(1 - \epsilon)$ is the binary thermodynamic entropy (see Sec. V of the Supplemental Material [42]).

Summary and outlook.—Landauer's principle implies a limit on the information throughput of irreversible computers as real hardware has finite power dissipation tolerance. Here, we derive a universal version of the finite-time Landauer principle that applies to any hardware implementation of the bit. It has been verified with energetic constraints and finite bit reset error thresholds. It would be interesting to ask if the finite-time Landauer principle can be extended beyond the detailed balance or, also, into the single-shot regime, i.e., statements about the work cost that hold in every single shot of an experiment. While we mentioned that there are reasons to think that quantum coherence does not allow one to surpass our bound, this should be probed further, e.g., using the set-ups of [12–18].

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*Corresponding author.

dahlsten@sustech.edu.cn

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